

CHM111 Practice Exam on chapter(s): 1, 2

Name: \_\_\_\_\_

These questions assess your understanding of the material from Chapters 1, 2 of Silberberg. Please write your name on the sheets of paper that you use and show all your work.

1. Write the chemical formula of strontium sulfide heptahydrate.  
ANSWER: \_\_\_\_\_
2. A nurse measures the weight of his patient's medicine. The weight is 0.0050 g. What is the weight in milligrams? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
3. Provide the chemical formula for the molecular compound carbon monoxide.  
ANSWER: \_\_\_\_\_
4. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is  $3.84 \times 10^{-5}$  g. What is the mass in micrograms? Report your answer in scientific notation.  
ANSWER: \_\_\_\_\_
5. A civil engineer measures the length of an airport runway. The length is 2271 m. What is the length in kilometers? Report your answer in scientific notation.  
ANSWER: \_\_\_\_\_
6. An optician measures the diameter of her patient's pupil. The diameter is 0.0018 m. What is the diameter in millimeters? Report your answer in scientific notation.  
ANSWER: \_\_\_\_\_
7. A geologist measures the weight of an ore sample. The weight is 4.8303 kg. What is the weight in grams? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
8. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 231 GHz. What is the frequency in hertz? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
9. A geophysicist measures the pressure in a borehole. The pressure is  $7.721 \times 10^7$  Pa. What is the pressure in megapascals? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
10. A botanist measures the thickness of a leaf. The thickness is 8.6 mm. What is the thickness in meters? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
11. Provide the chemical symbol of the most commonly formed ion of silver and indicate whether it is a cation or an anion.  
ANSWER: \_\_\_\_\_
12. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is  $5.1397 \times 10^{-7}$  m. What is the wavelength in nanometers? Report your answer as a decimal.  
ANSWER: \_\_\_\_\_
13. Write the name of the inorganic acid  $\text{H}_2\text{SO}_4$ .  
ANSWER: \_\_\_\_\_
14. Provide the name and empirical formula of the ionic compound that is formed by the  $\text{NH}_4^+$  cation and the  $\text{HSO}_3^-$  anion.  
ANSWER: \_\_\_\_\_
15. Write the chemical formula for arsenous acid.  
ANSWER: \_\_\_\_\_

16. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is  $7.49 \times 10^{-14}$  m. What is the wavelength in femtometers? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

17. A cell biologist measures the width of a cell. The width is 93  $\mu\text{m}$ . What is the width in meters? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

18. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 91 fs. What is the time in seconds? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

19. A physical chemist measures the width of a molecule. The width is 89 nm. What is the width in meters? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

20. Provide the name and empirical formula of an ionic compound that is formed by the hydronium cation and the sulfide anion.

ANSWER: \_\_\_\_\_

21. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$17.09 \frac{\text{mol}}{\text{L}} \times 0.824 \text{ L}$$

ANSWER: \_\_\_\_\_

22. Provide the element symbol that represents scandium.

ANSWER: \_\_\_\_\_

23. A millworker measures the width of a plank. The width is 6.4 dm. What is the width in meters? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

24. Answer the following questions: is  $\text{Be}^{2+}$  an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: \_\_\_\_\_

25. Sarah needs to know the mass of a sample of  $\text{Rb}_2\text{O}_2$  to 4 significant figures. She puts the sample on a digital scale. This is what the scale shows: 31.25 g. If this measurement is precise enough for Sarah, round it to 4 significant figures and report your answer in scientific notation.

ANSWER: \_\_\_\_\_

26. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is  $8.38 \times 10^{10}$  m. What is the distance in gigameters? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

27. A lab technician measures the weight of a reagent. The weight is 5486 g. What is the weight in kilograms? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

28. As part of her duties as an analytical chemist, Patricia measures the amounts of elements  $E_1$  and  $E_2$  found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than  $E_1$  and  $E_2$ . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element  $E_1$  that Patricia would find in a new 10.0-g sample of X. Round your answer to two significant figures.

| sample | mass of $E_1$ | mass of $E_2$ |
|--------|---------------|---------------|
| 1      | 11.7 g        | 6.5 g         |
| 2      | 7.8 g         | 4.3 g         |
| 3      | 11.2 g        | 6.2 g         |
| 4      | 9.5 g         | 5.3 g         |

ANSWER: \_\_\_\_\_

29. A geologist measures the height of a mountain. The height is 7.3841 km. What is the height in meters? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

30. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$950. \text{ mol} \times 21.54 \text{ L}$$

ANSWER: \_\_\_\_\_

31. A shipwright measures the length of a bolt. The length is 64.0 cm. What is the length in meters? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

32. A nuclear chemist measures the mass of an artificial element. The mass is 54 pg. What is the mass in grams? Report your answer as a decimal.

ANSWER: \_\_\_\_\_

33. Determine whether potassium and iodine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form.

ANSWER: \_\_\_\_\_

34. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$16.3112 \text{ g} - 2.1269 \text{ g}$$

ANSWER: \_\_\_\_\_

35. Benjamin must note the concentration of a solution of KF in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Benjamin weighs an empty graduated cylinder and finds its mass to be 7.07 g. He then puts some solid KF into the graduated cylinder and weighs it. With the KF added, the cylinder weighs 78.27 g. He adds water to the graduated cylinder and dissolves the KF completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 66.02 mL. What concentration should Benjamin write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: \_\_\_\_\_

36. Provide the name for the molecular compound  $\text{NF}_3$ . If the compound has a common name, you may provide that instead.

ANSWER: \_\_\_\_\_

37. As part of her duties as an analytical chemist, Ashley analyzes a sample of an unknown pure substance X and finds that it contains 5.8 g of element E<sub>1</sub> and 5.0 g of element E<sub>2</sub>, and no other elements. She then analyzes three other substances that also contain only elements E<sub>1</sub> and E<sub>2</sub> and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance?

| sample | mass of E <sub>1</sub> | mass of E <sub>2</sub> |
|--------|------------------------|------------------------|
| X      | 5.8 g                  | 5.0 g                  |
| A      | 29.2 g                 | 4.2 g                  |
| B      | 47.4 g                 | 13.8 g                 |
| C      | 64.5 g                 | 13.9 g                 |

ANSWER: \_\_\_\_\_

38. A pharmacist measures the mass of a powdered drug. The mass is 1411 mg. What is the mass in grams? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

39. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$12.49 \text{ g} + 6.641 \text{ g}$$

ANSWER: \_\_\_\_\_

40. Provide the number of protons and the number of neutrons there are in an atom of vanadium-51.

ANSWER: \_\_\_\_\_

41. Provide the name of the element that the element symbol Fe represents.

ANSWER: \_\_\_\_\_

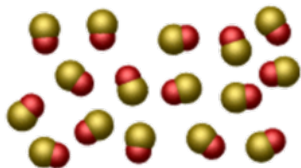
42. Classify PN as either an ionic or a molecular compound.

ANSWER: \_\_\_\_\_

43. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.305 m. What is the thickness in centimeters? Report your answer in scientific notation.

ANSWER: \_\_\_\_\_

44. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture?



ANSWER: \_\_\_\_\_

45. An engineer measures the capacity of an electricity-generating plant. The capacity is 493 MW. What is the capacity in watts? Report your answer as a decimal.

ANSWER: \_\_\_\_\_